1 Kimodo-based Motion Generation and G1 Simulation

The Kimodo branch investigates how text-conditioned motion generation can support the project's humanoid tabletop manipulation task. Its objective is to connect generated body motion to executable G1 references, coordinate the fingers during grasping, and assess the resulting interaction in the shared simulation environment. The work spans baseline reproduction, deployment on Moore Threads hardware, robot-interface development, physical evaluation, and data preparation for the bonus task. Figure 1 summarizes this complete branch. We use released Kimodo and SONIC models; their pretraining and controller design are prior work, while the deployment, adaptation and evaluation described below are project work.[2, 3]

1.1 Baseline reproduction and environment validation

We first established a functional baseline in the existing Moore Threads S4000 environment, preserving the working Python, PyTorch and MUSA framework. Checks covered executable and model paths, device availability, checkpoint loading, language-model adapters, generated motion validity, and export. The earlier workspace retains a SOMA checkpoint, a walking-motion output and a preview video. The subsequent G1 baseline used the released `Kimodo-G1-RP-v1` checkpoint and a simple instruction, "A person raises the right hand," to generate 60-frame and 150-frame motions at 30 Hz. Saved motion files and visual previews established that text input, inference and export were operational. This is functional reproduction of the released pipeline, rather than reproduction of the paper's quantitative benchmark.[1]

Deployment reused Python 3.10, PyTorch 2.9 and `torch_musa`. Compatibility work addressed the Llama-based LLM2Vec loading sequence and device placement in segmented constraint processing. The text encoder uses a Llama-3-8B-Instruct-MNTP base with a supervised LLM2Vec adapter. In the inspected grasping configuration, the motion generator ran on MUSA and the frozen generic text encoder ran on the CPU. Thus, investigating text-encoding compatibility on MUSA should not be confused with the device used in the evaluated pipeline. Environment notes, reproduction instructions, saved outputs and presentation material documented the progression from a working demo to the robot experiments. A separate grasping evaluation was necessary because a valid motion file alone does not establish physical object manipulation.

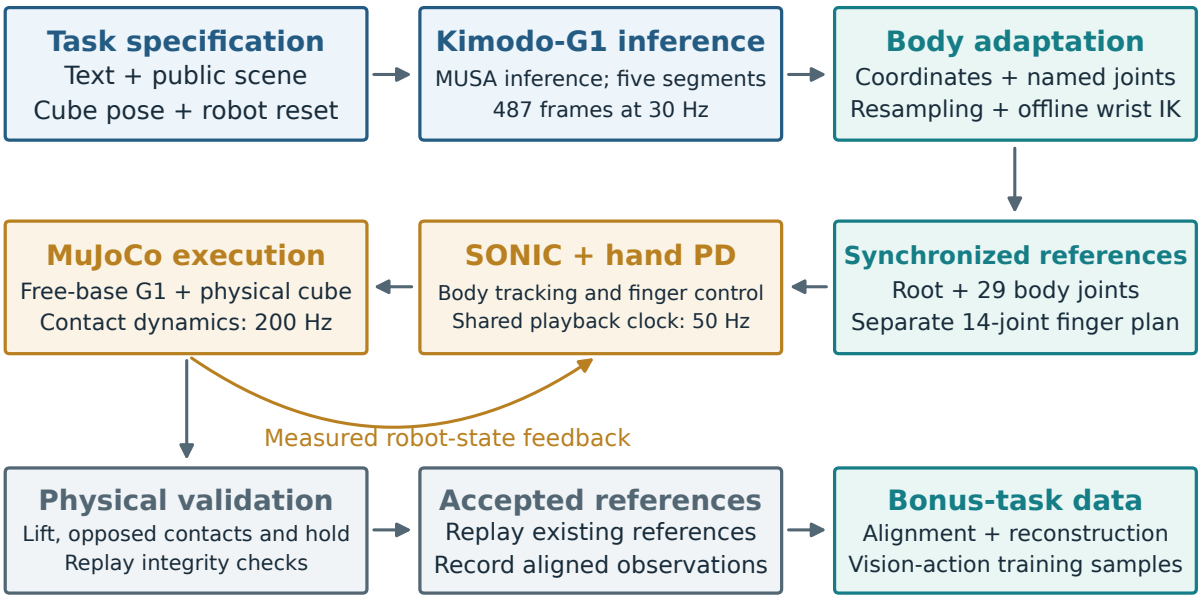


Figure 1: The team's Kimodo branch, from task specification to validated data. The finger reference is constructed separately from the generated body motion. Measured-state feedback acts at the execution layer; Kimodo does not regenerate motion online during replay. Data production replays accepted references and applies additional checks.

1.2 From generated motion to executable G1 references

Task-conditioned generation. The grasping sequence was organized into five text segments describing raising the hand, approaching the cube, grasping, lifting and holding. Their source lengths were 120, 120, 60, 60 and 127 frames. Public scene geometry, the cube pose, the robot reset and task constraints provided structured conditioning. One SDK entry invoked five internal sampling stages; these stages are successive parts of one motion, not five candidates from which a successful trial is selected. The historical fixed-scene batch used 100 denoising steps per segment and text/constraint guidance weights of 2 and 4. These settings describe that batch; later development configurations are not pooled with it.

Representation, coordinates and timing. Kimodo's 34-node G1 skeleton contains the pelvis, 29 hinge-joint nodes and four auxiliary nodes. These are different quantities from the robot's 29 actuated body joints. The converter exports 36 generalized position coordinates: three root translations, four quaternion components and 29 body angles. We checked joint names against the canonical robot model and reordered the angles to the SONIC convention. Root pose conventions, initial-frame alignment, forward kinematics and velocity consistency were checked before replay. Figure 2 separates this representation conversion from temporal resampling.

The coordinate interface maps a MuJoCo position (x, y, z) to Kimodo coordinates (y, z, x), together with the corresponding orientation transformation. An inverse constraint conversion and matrix-preserving pose codec maintain agreement between task conditioning and exported motion. Linear interpolation converts positions and body angles from 30 to 50 Hz, while spherical interpolation (SLERP) resamples root quaternions in `wxyz` order. The 487 source frames span 16.20 s; the final 810-frame reference spans timestamps 0–16.18 s. Nominal physical playback comprises 3,240 steps at 200 Hz, or 16.20 s. Reference metadata records joint names, units and time conventions; checkpoint/source hashes, configuration snapshots and retained failure records support traceability.

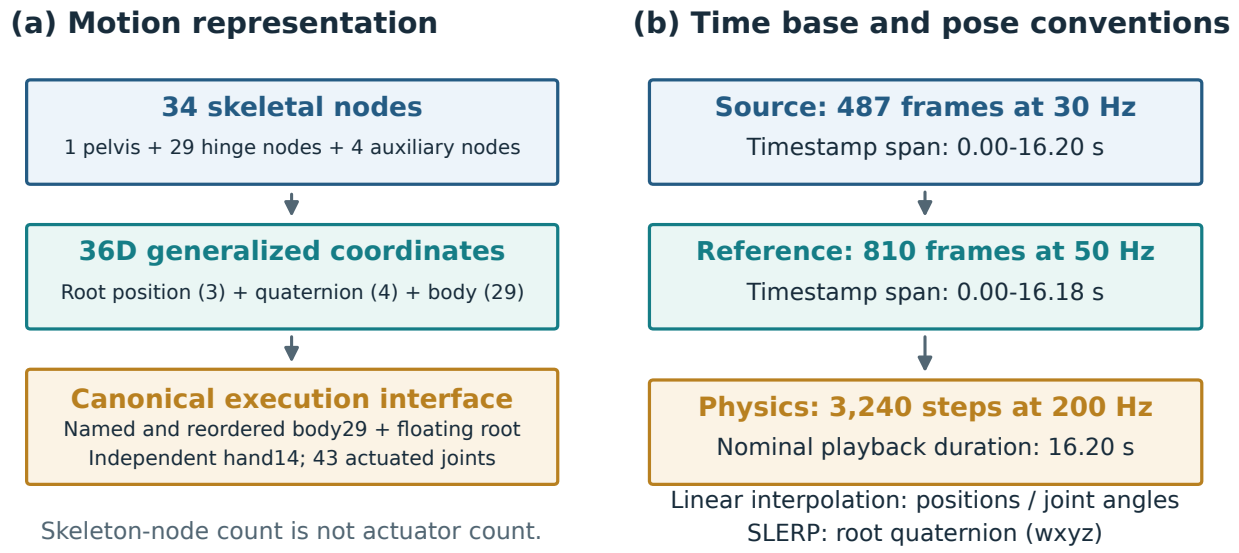


Figure 2: Spatial and temporal interfaces. Skeleton nodes, generalized coordinates and actuated joints are explicitly distinguished. The additional 14 finger joints are generated by the hand-motion interface. The difference between reference timestamp span and playback duration follows from discrete sampling.

Offline adaptation and hand coordination. The evaluated system includes task-specific adaptation beyond format conversion. In the historical fixed-scene configuration, the adapter constrains the generated wrist motion within a task-space corridor and solves the seven right-arm joints through inverse kinematics. The other body references remain at the canonical neutral pose. That configuration also uses a fixed tracking correction obtained from its own Kimodo development replay. Accordingly, grasp performance belongs to the combined generator, adapter and controller, rather than to unmodified model output.

Kimodo does not directly provide the 14 Dex3 finger-joint commands used here. A separate schedule generates seven joints per hand, coordinating preclosure, closure, lifting and holding with the body reference. Both reference streams share a playback index. SONIC tracks the body at 50 Hz, and the canonical hand PD

controller regulates the fingers during 200 Hz MuJoCo simulation. The simulated robot has a free base; a neutral root reference does not fix or weld the robot to the world.

1.3 Simulation setting and acceptance criteria

The task uses a 0.07 m cube with mass 0.10 kg on a table whose surface is 0.75 m above the floor. The fixed-scene collection places the cube center at $(0.40, -0.15, 0.785)$ m. The five-position experiment uses a separate case set, shown in Figure 3; its central case is not the collection placement. The canonical robot, scene, SONIC release and controller settings provide the shared execution basis.

The intended grasp criterion requires cube lift of at least 0.10 m, effective opposed contacts between the task-hand thumb and index or middle finger, contact forces above 10^{-4} N, and no cube–table contact throughout a continuous qualifying hold of at least 2.0 s. Complete finite playback and absence of falls, artificial support, welds or external assistance are also required. Hand–table contact is recorded diagnostically and is not by itself a task failure. Individual evaluators differ in whether they check geometric opposition, require terminal rather than maximum hold, and impose extra contact gates. The results below retain these distinctions instead of treating every stored success flag as the same metric.

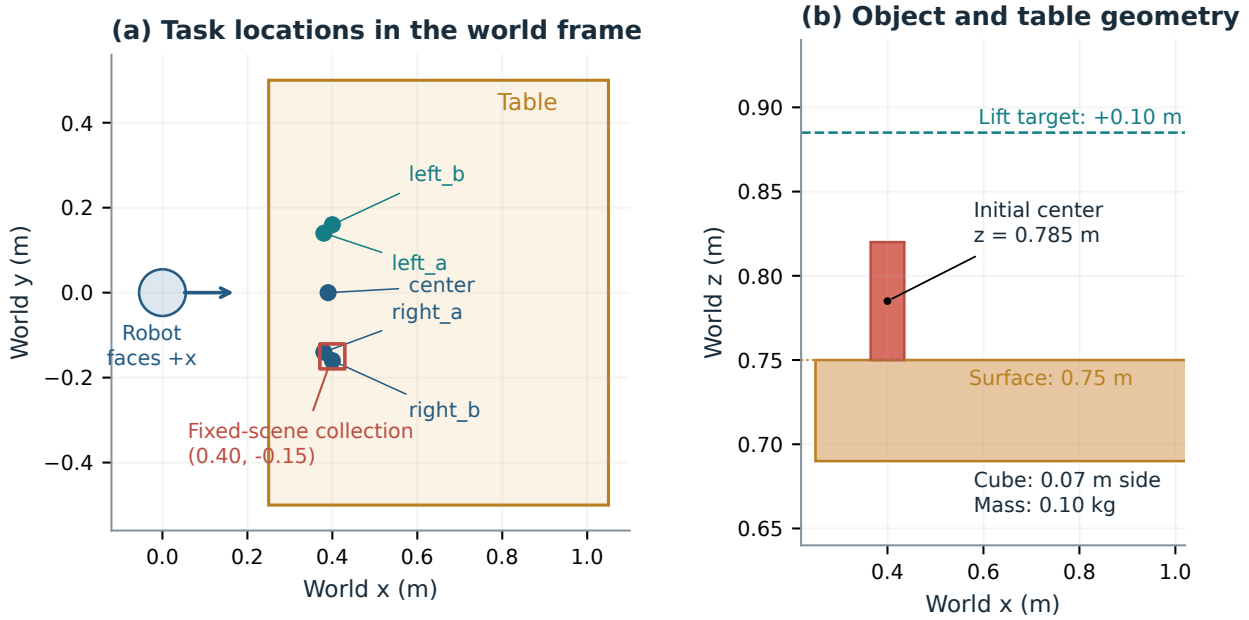


Figure 3: Scene geometry and tested placements. The table occupies $x = 0.25$–1.05 m and $y = -0.50$–0.50 m. The five-position test uses left-arm cases at $(0.38, 0.14)$ and $(0.40, 0.16)$, a right-arm center case at $(0.39, 0)$, and right-arm cases at $(0.38, -0.14)$ and $(0.40, -0.16)$ m. The red outline identifies the distinct fixed-scene collection placement. Right-panel axis scales differ.

1.4 Experimental observations and their scope

Fixed-scene repeatability. The historical collection delivered 111 validated references from 120 observed trial starts, a delivery yield of 92.5%. Its ledgers contain 119 entries; the additional start has no final ledger verdict. Seven starts stopped before replay because DDS was occupied, one stopped during replay, and one remained unfinalized before replay. These outcomes remain in the delivery denominator. They are not measured zero-lift physical failures. The 111 trials with final physical verdicts all passed, but this conditional subset must not replace the all-start denominator. Successful trials reached a mean maximum lift of 0.1261 ± 0.0022 m, while generation time across 119 recorded trials was 51.280 ± 1.435 s (sample standard deviations).

Figure 4 shows actual saved measurements and rendered states from historical seed 100. It reached 0.1230 m maximum lift and a geometrically verified 4.330 s hold. Its saved general replay timer reports 4.615 s under a different check; the figure deliberately uses the independent geometric-validation interval. No new generation or simulation was performed to produce this illustration.

Position transfer. The October 8 comparison completed five Kimodo cases with zero successful cases under its combined acceptance gates. It adapted a cached reference using deformation, mirroring and IK, with up to four replays and two deterministic corrections per case. It therefore measures transfer through that adaptation procedure rather than fresh single-sample generation at each location. Across the five final failed replays, maximum cube lift was only 5.19–14.20 mm and qualifying hold was zero. The experiment also used a terminal-hold rule, unlike the historical collection's longest-interval rule. The shared cross-method results belong in the overall experimental comparison; these Kimodo results establish a limitation of the tested transfer method.

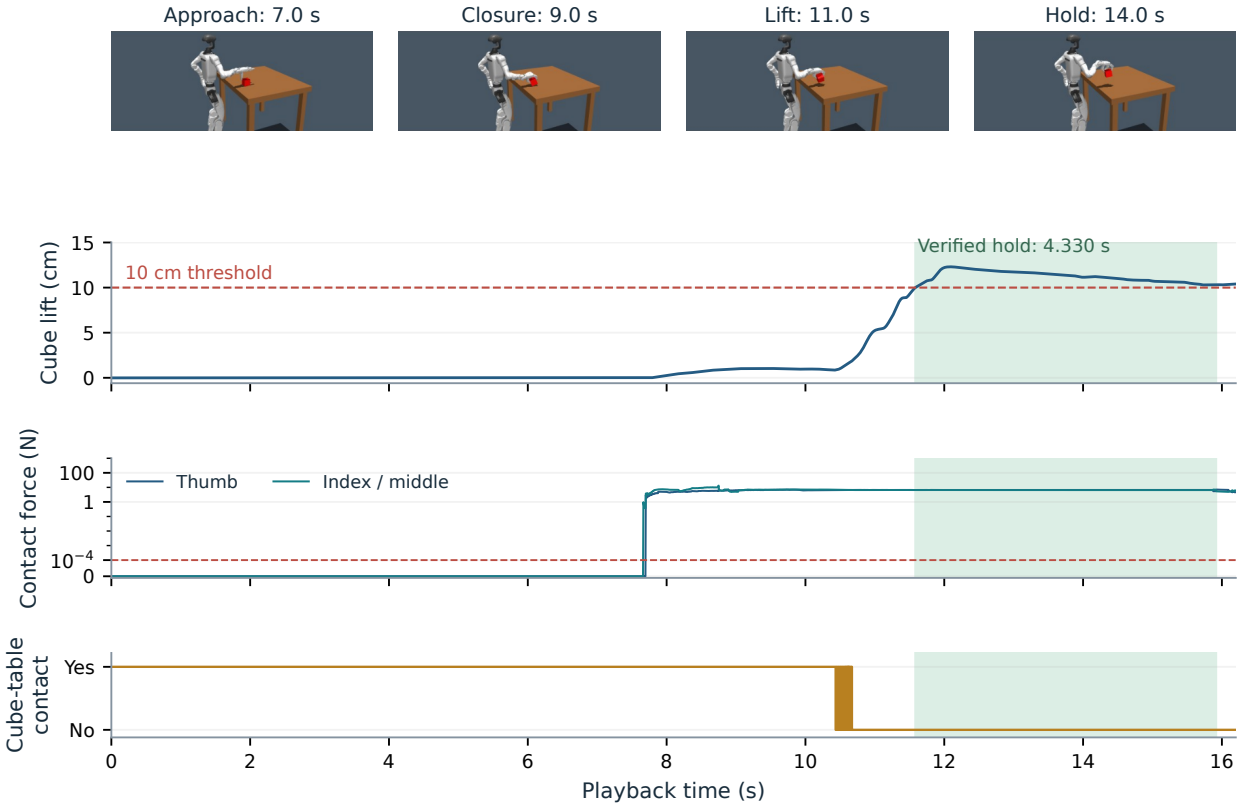


Figure 4: Measured evidence from the historical fixed-scene seed-100 replay. The top row contains cropped views from an existing rendering of the saved simulation states. Curves use all 3,240 original playback physics steps, with time relative to playback start. Dashed lines show the lift and force thresholds; the force axis uses a symmetric-log scale. Green shading marks the independent geometric validator's 4.330 s interval, rather than the broader 4.615 s replay timer. Cube–table contact is absent during that interval. This representative episode is not evidence of success at the five other placements.

Evaluation scope. The October 16 attribution study is under revision to align its acceptance implementation with the agreed task definition. In particular, hand–table contact alone must not be counted as a grasp failure. Pending finalized validation, this study is excluded in full from the reported grasp-success statistics, including both provisional positive and negative flags and its ablation result. Earlier completed experiments retain their original scope and denominators. The saved development records are preserved separately; no final success or failure is inferred from the work in progress.

Information independence. Early development replay 07 used ARDY-informed grasp geometry and is excluded from independent Kimodo evaluation. The later historical collection records identify Kimodo's own development replay as the source of tracking compensation. The inspected source chain records no ARDY grasp trajectory, outcome or grasp-derived calibration as an input to that branch. Shared public scene assets and the generic text encoder are disclosed separately. This evidence concerns the recorded source chain, rather than every possible historical design influence, and does not make bounded-retry development a strict single-shot qualification experiment. The October 16 study will be described using its finalized configuration and validation records once those are available.

Table 1: Kimodo outcomes kept separate by experimental unit and protocol. Counts are not pooled into one success rate.

Experiment	Saved outcome	Interpretation
Historical fixed-scene collection	111 valid deliveries / 120 starts	Pipeline delivery yield; nine starts have no complete final physical verdict.
October 8 position transfer	0 / 5 accepted cases	Cached-reference adaptation with bounded corrections and retries.
October 16 attribution study	Under revision; not included	Both provisional positive and negative outcomes await finalized validation.
October 16 bonus-data example	1 accepted export, 161 samples	Reused generated motion after an adapter change; not a new independent sampling trial.

1.5 Bonus-task data and practical lessons

Accepted references feed the project's shared vision–action recording pipeline. Images are rendered from recorded states, and explicit identifiers join observations, decisions, commands and physics records. Each sample contains a 320×240 RGB image at 10 Hz, three state observations with a validity mask, and five future targets at 50 Hz control offsets 2–6 (40–120 ms). Each target combines a 64-dimensional post-FSQ SONIC token with 14 absolute finger-joint angles. These are SONIC supervision tokens, not Kimodo model tokens. Figure 5 illustrates the alignment.

A current accepted example, `kimodo_fresh_20261016_03`, contains 162 RGB frames and 161 exported samples, with body-target shape $161 \times 5 \times 64$ and hand-target shape $161 \times 5 \times 14$. Its data-production replay reports 0.1200 m maximum lift and 4.565 s hold. This run reused the preceding run's raw generated motion after changing clearance; it is evidence of a successful data-export path, not an additional fresh model sample. The earlier inspected 17-episode recording batch had yielded 16 accepted datasets and 2,576 samples, but its original directory was unavailable during the latest preparation of this section. That historical snapshot is therefore not used as a current inventory claim.

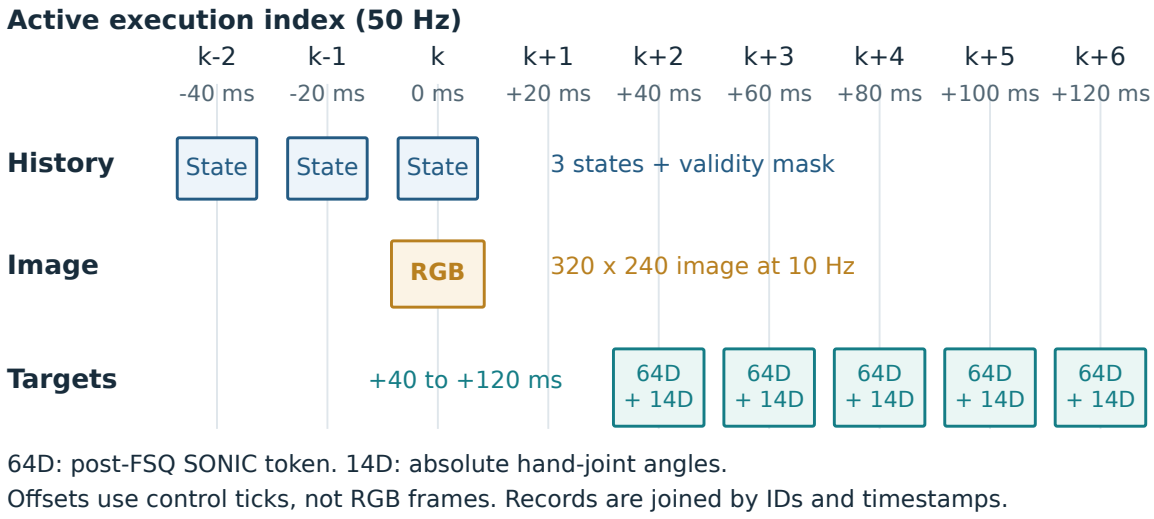


Figure 5: Vision–action sample alignment. Index k denotes the active control sequence, not the image number or an absolute record identifier. History uses $k-2, k-1, k$, with startup masking; future labels use $k+2$ through $k+6$. The nominal offsets are based on 50 Hz control. Asynchronous records are matched using saved IDs and timestamps.

The practice demonstrates that baseline reproduction, executable-reference construction, physical grasping and training-data acceptance are separate milestones. Coordinate consistency and finger synchronization are necessary for execution, while grasp reliability also depends on the task adapter and contact geometry. The experiments support successful operation in particular fixed settings, expose limited transfer in the tested cached-reference procedure, and reveal why evaluator consistency matters. Future comparisons should use matched case sets, durations, development budgets and acceptance checks, with immutable records of the settings actually executed. The delivered data establishes validated supervision; it does not by itself demonstrate a trained downstream policy or improved policy performance.

References

[1] NVIDIA. *Kimodo: Scaling Controllable Human Motion Generation*. Technical report and project, 2026. https://research.nvidia.com/labs/sil/projects/kimodo/.

[2] NVIDIA. *Kimodo-G1-RP-v1: Model Card*. https://huggingface.co/nvidia/Kimodo-G1-RP-v1.

[3] NVIDIA. *GR00T WholeBodyControl: SONIC Release and Deployment Code*. https://github.com/NVlabs/GR00T-WholeBodyControl.